# EE5341: MOS Device Modelling & Characterization

## Verilog A – Assignment 2

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## Problem Statement

Implement the full wave bridge rectifier using the Verilog-A compact model for diode.

## Verilog-A Code:

```
`include "constants.vams"
`include "disciplines.vams"

module diode_basic(anode,cathode);
    inout anode,cathode;
    electrical anode, cathode;

    //Parameters
    parameter real Is =1e-14 from (0:inf); //Saturation current
    parameter real Temp =27.0;             //Temperature

    real Vt; //Thermal voltage
    real Vd; //Voltage across diode
    real Id; //Diode current

    analog begin

        //Thermal voltage: Vt =kT/q
        Vt =`P_K*(Temp + `P_CELSIUS0) /`P_Q;
        Vd = V(anode,cathode);
        Id = Is* (limexp(Vd/(Vt)) -1.0);
        I(anode,cathode) <+ Id;
    end
endmodule
```

Figure 1: Verilog-A module implementation for the diode (`diode_basic.va`).

## Circuit Setup & Simulation Details:

- **Input Voltage Source ($V_1$):** AC voltage source with amplitude $U = 5\,\mathrm{V}$ and frequency $f = 50\,\mathrm{Hz}$.
- **Diode Models:** Four diodes ($D_1, D_2, D_3, D_4$) modeled using the Verilog-A module configured in a full-wave bridge topology.
- **Load Resistor ($R_1$):** $1000\,\Omega$ connected across terminal nodes `OUT` and `V_1`.

- **Transient Simulation** ($TR1$)**:** Stop time = 60 ms, Total calculation points = 501.

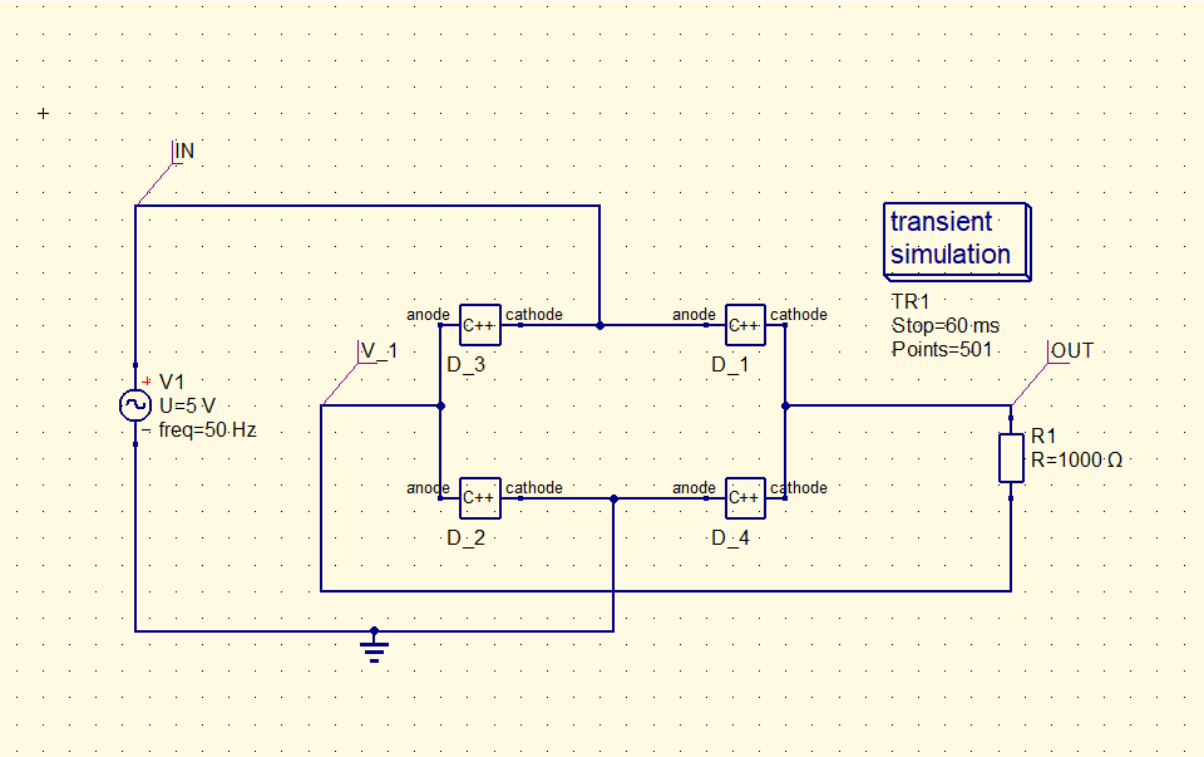



Figure 2: Full-Wave Bridge Rectifier schematic

# Plots Obtained:

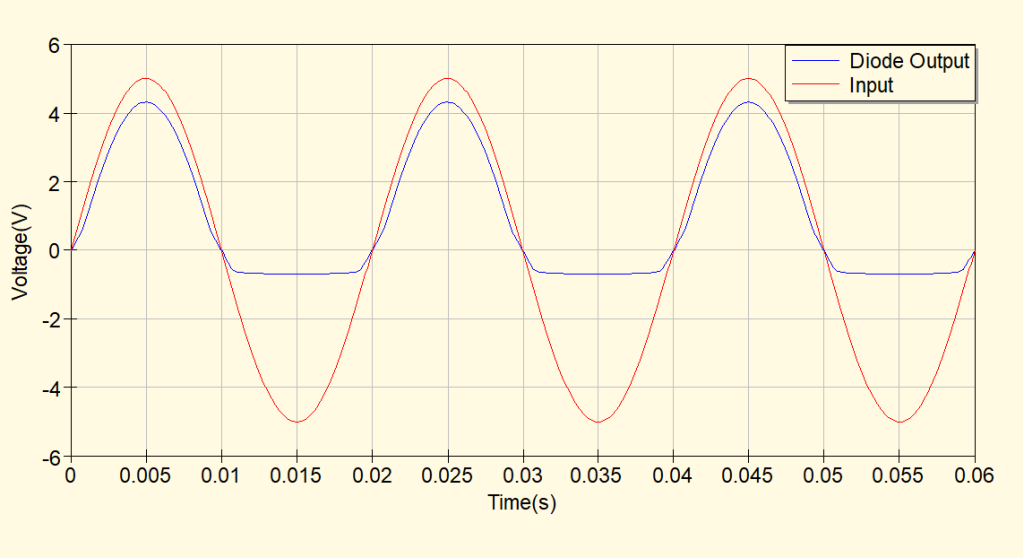



Figure 3: Transient Simulation Plot showing Input AC voltage **IN.Vt** and Single-Node Diode Output voltage **OUT.Vt** as a function of time.

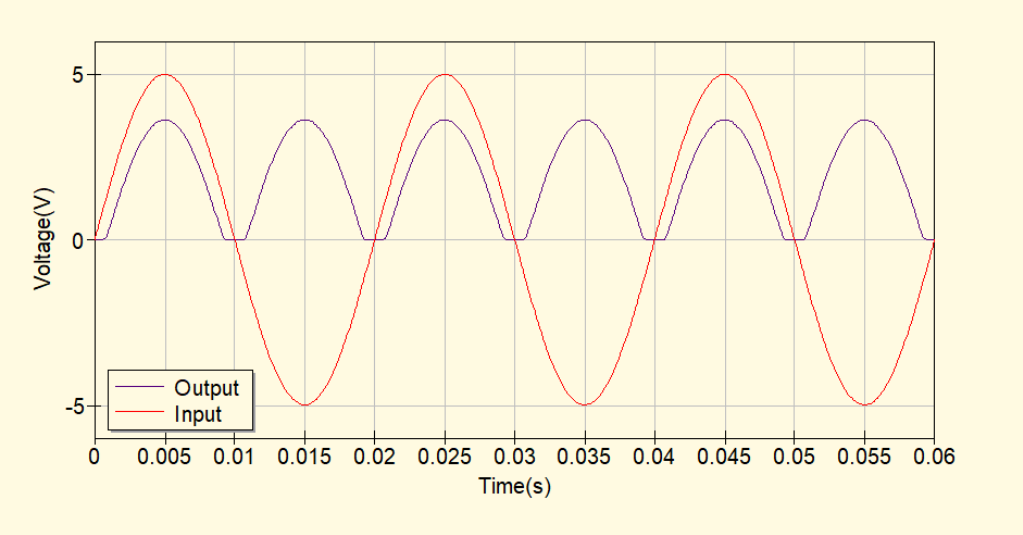



Figure 4: Transient Simulation Plot showing Input AC voltage **IN.Vt** ($5\,\text{V}, 50\,\text{Hz}$) and rectified differential load voltage **OUT.Vt** − **V_1.Vt** as a function of time.

# Inference:

- **Full-Wave Rectification:** Both positive and negative half-cycles of the 50 Hz sinusoidal input waveform ($V_{peak} = 5\,\text{V}$) are converted into unidirectional positive pulses across load resistor $R_1$, yielding a ripple frequency of 100 Hz ($T_{ripple} = 10\,\text{ms}$).
- **Diode Drop Impact ($2V_D$):** The peak differential output voltage reaches approximately 3.7 V, which accounts for the combined threshold voltage drop across two conducting diodes in series ($2V_D \approx 1.3\,\text{V}$) during each conduction phase.
- **Single Diode Output Characteristics:** The single-node output voltage at node `OUT` peaks at $+4.35\,\text{V}$ ($V_{peak} - V_D$) during forward conduction of diode $D_1$ in the positive half-cycle, and clamps at $-0.65\,\text{V}$ (one diode drop below ground) during the negative half-cycle due to diode $D_4$ conduction.
- **Conduction Cycles:** Diodes $D_1$ and $D_2$ conduct during the positive half-cycles of $V_1$, while $D_3$ and $D_4$ conduct during the negative half-cycles, maintaining consistent current direction through $R_1$.